Diagnostic Summary Report

Patient Overview

- **Patient Name:** Ashley Grahm
- **Age:** 20
- **Gender:** Female
- **Primary Complaint:** Stomach pain and vomiting

Red-Flag Alert

- **Hypoxia:** SpO■ = 91 % (below normal, suggests significant dehydration or early respiratory compromise)
- **Leukocytosis:** WBC = 10,570 /µL (possible infection or early sepsis)
- **New■onset diabetes indicators:** Fasting glucose = 141 mg/dL, HbA1c = 7.1 % (risk of hyperglycemia■related dehydration)
- **Persistent vomiting:** Increases risk of electrolyte imbalance and volume depletion

Risk Stratification

- **Level:** High
- **Key Drivers:**
- Mid■abdominal sharp pain worsened by movement
- Low■grade fever (100.1 °F)
- Leukocytosis (WBC 10,570)
- Recent fast■food burger ingestion
- Hypoxia (SpO■ 91 %) likely secondary to dehydration
- Elevated fasting glucose and HbA1c indicating new■onset diabetes

Probable Diagnosis

- **Condition:** Acute food■borne gastroenteritis (bacterial) likely secondary to recent fast■food burger consumption
- **Confidence:** 70 %
- **Justification (Explainability Pack):**
- The patient’s acute mid■abdominal pain, vomiting, low■grade fever, and leukocytosis align with classic bacterial gastroenteritis.
- Recent ingestion of a fast■food burger provides a plausible exposure source.
- Hypoxia is interpreted as a consequence of dehydration from vomiting.
- Elevated glucose/HbA1c suggest underlying diabetes that can exacerbate infection and fluid loss.
- **Reasoning:** The constellation of gastrointestinal symptoms, inflammatory markers, and exposure history points toward an infectious etiology rather than a primary surgical abdomen.
- **Supporting Evidence:**
- Mid■abdominal sharp pain worsened by movement, present since this morning
- Vomiting with low■grade fever (100.1 °F)
- Leukocytosis (WBC 10,570)
- Recent ingestion of a fast■food burger
- Hypoxia (SpO■ 91 %) likely from dehydration

Differential Diagnoses

- **Early appendicitis** – Pain could evolve to RLQ; however, no localized tenderness, rebound, or migration reported.
- **Viral gastroenteritis** – Similar presentation, but leukocytosis and bacterial exposure favor bacterial cause.
- **Acute pancreatitis** – Epigastric pain/vomiting possible; no back radiation or elevated lipase noted.
- **Pyelonephritis** – No flank tenderness or urinary symptoms.
- **New■onset diabetes with hyperglycemia■related dehydration** – Elevated glucose/HbA1c; glucose not high enough for DKA yet contributes to dehydration.

Key Laboratory Findings

- **WBC Count:** 10,570 /µL (High) – supports infection/inflammation.

- **Fasting Blood Sugar:** 141 mg/dL (High) – indicates hyperglycemia/possible new diabetes.
- **HbA1c:** 7.10 % (High) – confirms diabetes.
- **Vitamin B12:** < 148 pg/mL (Low) – deficiency.
- **IgE:** 492.30 IU/mL (High) – suggests allergic/atopic tendency.
- **Homocysteine:** 23.86 µmol/L (High) – cardiovascular risk marker.
- **25■OH Vitamin D:** 8.98 ng/mL (Low) – deficiency.
- **Triglyceride:** 168 mg/dL (High) – mild hypertriglyceridemia.
- **Direct LDL:** 100.39 mg/dL (High) – borderline elevated LDL.

Recommended Plan

Required Investigations

| Test | Purpose / What it Confirms | |------|---------------------------| | CBC with differential | Verify leukocytosis, assess anemia or neutrophil shift | | Comprehensive metabolic panel | Electrolytes, renal/hepatic function, glucose trends | | Serum lactate | Detect tissue hypoperfusion or early sepsis | | Arterial blood gas | Evaluate oxygenation, acid■base status, rule out respiratory compromise | | Serum ketones (β■hydroxybutyrate) | Exclude early diabetic ketoacidosis | | Stool culture, ova & parasites, C. difficile PCR | Identify bacterial, parasitic, or C. diff infection | | Rapid viral gastroenteritis panel | Detect common viral pathogens (norovirus, rotavirus) | | Abdominal ultrasound | Screen for appendicitis, gallbladder disease, other intra■abdominal pathology | | Contrast■enhanced CT abdomen/pelvis (if US inconclusive) | Detailed assessment of intra■abdominal organs and possible complications | | Chest X■ray | Investigate cause of hypoxia (e.g., atelectasis, pneumonia) | | Pregnancy test (urine hCG) | Rule out pregnancy■related considerations (important for imaging & meds) | | Blood cultures (if febrile or septic picture) | Detect bacteremia/sepsis | | Serum lipase | Exclude acute pancreatitis | | Repeat SpO■ with supplemental O■ trial | Confirm improvement with oxygen therapy |

Suggested Medications / Treatments

- **Ondansetron 4–8 mg IV/PO q8h** – control nausea/vomiting.
- **IV normal saline bolus 1–2 L** (adjusted for weight & dehydration status) – rehydrate.
- **Acetaminophen 650 mg PO q6h PRN** – antipyretic/analgesic if no contraindication.
- **Empiric oral ciprofloxacin 500 mg BID for 3 days** *(consider only if high■risk bacterial gastroenteritis suspected and pending stool culture)*.
- **Short■acting insulin sliding scale** *(if glucose >180 mg/dL, monitor closely)*.
- **Supplemental oxygen** (nasal cannula 2 L/min) – maintain SpO■ > 94 %.

Medication disclaimer: A qualified human doctor must make the final prescribing decision after a physical exam and review of all test results.

Information Gaps

- Results of stool studies and abdominal imaging to confirm infectious vs. surgical etiology.
- Arterial blood gas and lactate values to assess severity of hypoxia and possible sepsis.
- Serum ketones and repeat glucose to rule out evolving diabetic ketoacidosis.
- Detailed physical exam findings (peritoneal signs, rebound tenderness, bowel sounds) that could modify urgency.